🌍 Climate Analytics – Capstone Project Report

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1. Introduction

This project focuses on analyzing global climate change data — including temperature anomalies, CO_2 emissions, rainfall, renewable energy adoption, and extreme weather events — to derive actionable insights. The workflow integrates Excel → SQL → Python → Data Science → Power BI, ensuring a complete pipeline from raw data cleaning to predictive modeling and dashboard storytelling.

Objective:

Predict temperature anomalies, classify regional climate risk, and identify vulnerable regions using multi-stage analytics.

2. Data Sources

- Temperature anomalies (2000–2025)
- CO_2 emissions (2000–2025)
- Rainfall records (2000–2025)

3. Methodology

Excel Cleaning

- Imported raw datasets into Excel.
- Cleaned missing values using Power Query.
- Applied conditional formatting to highlight regions with >2℃ anomaly.
- Built pivot tables for year-wise CO2 emissions.
- Created trendline charts and moving averages for rainfall.
- Classified regions into **Low/Medium/High risk** using nested IF functions.
- Exported cleaned datasets for SQL integration.

SQL Analysis

- Created tables for Temperature, CO2, Rainfall.
- Joined datasets on **Year + Region**.
- Calculated average anomalies per continent using GROUP BY.
- Applied window functions for cumulative CO2 emissions.
- Identified **Top 5 regions with highest anomalies**.
- Categorized emission levels using CASE WHEN (Low, Medium, High, Critical).
- Built stored procedure for yearly climate summary.
- Created a view for renewable energy adoption trends.

- Exported results into CSV for Python analysis.

Python Wrangling & Visualization

- Merged SQL-exported datasets into a single DataFrame.
- Verified no missing values or duplicates.
- Created scatter plots (CO_2 vs Temperature anomalies).
- Generated continent-level summaries (Asia had highest CO_2 emissions at ~3708 MtCO_2; Europe & North America showed highest anomalies ~1.1℃).
- Built regression model ($R^2 \approx 0.63$).
- Forecasted CO_2 emissions (ARIMA: 2026–2030 rising from ~1433 to ~1519 MtCO_2).
- Forecasted temperature anomalies (ARIMA: 2026–2030 rising from ~1.6℃ to ~1.81℃).
- Created rainfall anomaly heatmaps.
- Automated pipeline with functions for loading, cleaning, merging, and visualization.

Data Science Modeling

- Defined problem: Predict temperature anomalies.
- Train/Test split: 80/20.
- **Random Forest Regressor**: RMSE = 0.21℃, $R^2 = 0.86$ (best performance).
- **Linear Regression**: RMSE = 0.34℃, $R^2 = 0.63$.
- Feature importance:
    - Year (57.9%)

- CO_2 emissions (19%)
- Rainfall (19%)
- Region (4%)

- **Clustering (K-Means)**:
  - Cluster 0: Moderate Warm (Russia, Canada, Germany)
  - Cluster 1: Cool Low-CO_2 (South Africa, Argentina, Egypt)
  - Cluster 2: Hot High-CO_2 (USA, China)
  - Cluster 3: Wet Tropical (Brazil, Japan, Nigeria)

- **Classification Model**: Risk level prediction accuracy = 87.5%.

- **Vulnerability Analysis**: -
  - Most vulnerable countries: Russia (Avg anomaly 1.70℃, Max 2.95℃), Canada (1.50℃), China (1.14℃), USA (1.07℃).

  - Most vulnerable regions: Europe (Avg anomaly 1.11℃, 70.9% high-risk years), North America (1.10℃, 64.1% high-risk years).

Power BI Dashboard

- Imported processed datasets and model outputs.
- Built interactive maps for anomalies.
- Line charts for CO_2 vs Temperature.
- Forecast visuals for next 10 years.
- Cluster visuals for region segmentation.
- Risk classification cards.
- Storytelling page summarizing climate impact.
- Published dashboard as final deliverable.

4. Key Insights

- **Temperature anomalies are rising steadily**, with Europe and North America most affected.
- **CO_2 emissions strongly correlate with anomalies**, especially in USA and China.
- **Random Forest models outperform Linear Regression**, highlighting non-linear climate dynamics.
- **Clusters reveal distinct climate profiles** — e.g., Wet Tropical vs Hot High-CO_2.
- **Forecasts predict anomalies reaching ~1.8℃ by 2030**, intensifying global risks.

Conclusion

- **This capstone demonstrates a complete end-to-end climate analytics pipeline. By integrating multiple tools (Excel, SQL, Python, ML, Power BI), the project provides predictive insights and visual storytelling on climate change. The findings emphasize urgent action for regions with high anomalies and emissions, particularly Europe, North America, USA, China, and Russia.**